

EXP-4

EDA – DATA INSPECTION AND ANALYSIS

AIM

To inspect and analyze the dataset using DataFrame operations and calculate descriptive statistics such as mean, median, mode, range, variance, and standard deviation.

PROGRAM

```
import pandas as pd
```

```
data = {  
    'Gender': ['Male', 'Female', 'Male', 'Female', 'Male'],  
    'Married': ['Yes', 'No', 'Yes', 'Yes', 'No'],  
    'Dependents': ['0', '1', '3+', '2', '0'],  
    'Education': ['Graduate', 'Graduate', 'Not Graduate', 'Graduate', 'Graduate'],  
    'Self_Employed': ['No', 'Yes', 'No', 'No', 'Yes'],  
    'ApplicantIncome': [5720, 3076, 5000, 2340, 3276],  
    'CoapplicantIncome': [0, 1500, 2000, 2546, 0],  
    'LoanAmount': [110, 126, 208, 100, 78],  
    'Property_Area': ['Urban', 'Urban', 'Urban', 'Rural', 'Urban']  
}
```

```
df = pd.DataFrame(data)
```

```
print("First 5 Rows")
```

```
print(df.head())
```

```
print("\nLast 5 Rows")
```

```
print(df.tail())
```

```
print("\nData Information")
```

```
df.info()
```

```
print("\nDescriptive Statistics")
```

```
print(df.describe())
```

```
print("\nColumn Names")
```

```
print(df.columns)
```

```
categorical_cols = [
```

```
    'Gender',
```

```
    'Married',
```

```
    'Dependents',
```

```
    'Education',
```

```
    'Self_Employed',
```

```
    'Property_Area'
```

```
]
```

```
for col in categorical_cols:
```

```
    print(f"\nUnique values in {col}")
```

```
    print(df[col].unique())
```

```
df['Dependents'] = df['Dependents'].replace('3+', 3)
```

```
df['Dependents'] = pd.to_numeric(
```

```
    df['Dependents'],
```

```

errors='coerce'

)

print("\nValue Counts of Dependents")

print(df['Dependents'].value_counts())

cols = [

    'ApplicantIncome',

    'CoapplicantIncome',

    'LoanAmount'

]

for col in cols:

    print(f"\nStatistics for {col}")

    print("Mean:", round(df[col].mean(), 2))

    print("Median:", round(df[col].median(), 2))

    print("Mode:", df[col].mode().values)

    print("Range:", round(df[col].max() - df[col].min(), 2))

    print("Variance:", round(df[col].var(), 2))

    print("Standard Deviation:", round(df[col].std(), 2))

```

OUTPUT

```

First 5 Rows
  Gender  Married  Dependents  Education  Self_Employed  ApplicantIncome  \
0  Male      Yes           0   Graduate           No           5720
1  Female    No           1   Graduate           Yes           3076
2  Male      Yes          3+  Not Graduate           No           5000
3  Female    Yes           2   Graduate           No           2340
4  Male      No           0   Graduate           Yes           3276

```

```

  CoapplicantIncome  LoanAmount  Property_Area
0                0           110         Urban
1             1500           126         Urban
2             2000           208         Urban
3             2546           100         Rural
4                0            78         Urban

```

```

Last 5 Rows
  Gender  Married  Dependents  Education  Self_Employed  ApplicantIncome  \
0  Male      Yes           0   Graduate           No           5720
1  Female    No           1   Graduate           Yes           3076
2  Male      Yes          3+  Not Graduate           No           5000
3  Female    Yes           2   Graduate           No           2340
4  Male      No           0   Graduate           Yes           3276

```

```

  CoapplicantIncome  LoanAmount  Property_Area
0                0           110         Urban
1             1500           126         Urban
2             2000           208         Urban
3             2546           100         Rural
4                0            78         Urban

```

```

Data Information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4

```

```

Data columns (total 9 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Gender              5 non-null     object
1   Married             5 non-null     object
2   Dependents          5 non-null     object
3   Education            5 non-null     object
4   Self_Employed       5 non-null     object
5   ApplicantIncome     5 non-null     int64
6   CoapplicantIncome   5 non-null     int64
7   LoanAmount          5 non-null     int64
8   Property_Area       5 non-null     object
dtypes: int64(3), object(6)
memory usage: 492.0+ bytes

```

```

Descriptive Statistics
      ApplicantIncome  CoapplicantIncome  LoanAmount
count              5.000000             5.000000             5.000000
mean             3882.400000             1209.200000             124.400000
std              1416.227665             1164.183491              49.867825
min              2340.000000              0.000000              78.000000
25%              3076.000000              0.000000             100.000000
50%              3276.000000             1500.000000             110.000000
75%              5000.000000             2000.000000             126.000000
max              5720.000000             2546.000000             208.000000

```

```

Column Names
Index(['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed',
      'ApplicantIncome', 'CoapplicantIncome', 'LoanAmount', 'Property_Area'],
      dtype='object')

```

```

Unique values in Gender
['Male' 'Female']

```

```

Unique values in Married
['Yes' 'No']

```

```

Unique values in Dependents
['0' '1' '3+' '2']

```

```

Unique values in Education

```

```

Column Names
Index(['Gender', 'Married', 'Dependents', 'Education', 'Self_Employed',
      'ApplicantIncome', 'CoapplicantIncome', 'LoanAmount', 'Property_Area'],
      dtype=object)

Unique values in Gender
['Male' 'Female']

Unique values in Married
['Yes' 'No']

Unique values in Dependents
['0' '1' '3+' '2']

Unique values in Education
['Graduate' 'Not Graduate']

Unique values in Self_Employed
['No' 'Yes']

Unique values in Property_Area
['Urban' 'Rural']

Value Counts of Dependents
Dependents
0    2
1    1
3    1
2    1
Name: count, dtype: int64

Statistics for ApplicantIncome
Mean: 3882.4
Median: 3276.0
Mode: [2340 3076 3276 5000 5720]
Range: 3380
Variance: 2005700.8
Standard Deviation: 1416.23

Value Counts of Dependents
Dependents
0    2
1    1
3    1
2    1
Name: count, dtype: int64

Statistics for ApplicantIncome
Mean: 3882.4
Median: 3276.0
Mode: [2340 3076 3276 5000 5720]
Range: 3380
Variance: 2005700.8
Standard Deviation: 1416.23

Statistics for CoapplicantIncome
Mean: 1209.2
Median: 1500.0
Mode: [0]
Range: 2546
Variance: 1355323.2
Standard Deviation: 1164.18

Statistics for LoanAmount
Mean: 124.4
Median: 110.0
Mode: [ 78 100 110 126 208]
Range: 130
Variance: 2486.8
Standard Deviation: 49.87

```

RESULT

Thus, the given program was successfully executed. The dataset was inspected using DataFrame functions, unique values and value counts were identified, and measures of central tendency and dispersion were calculated successfully.